cs224n a3 mzx

June 8, 2026

(1)

a

i one batch only count $1 - \beta_1$ (usually 10%) in the update, so the update is less likely to swing the direction. so we can get a more stable convergence
ii $m/\sqrt{v}$ will be normalised to a suitable size. parameters with small v get larger updates (compared with pure sgd); params with small gradients can still have a suitable update speed. params with large gradients are scaled down can reduce the oscillation.

b

i

$$\gamma = \frac{1}{1 - p_{\text{drop}}}$$

$$\begin{aligned}
\mathbb{E}_{p_{\text{drop}}}[h_{\text{drop}}]_i &= \mathbb{E}_{p_{\text{drop}}}[\gamma d \odot h]_i \\
&= \mathbb{E}_{p_{\text{drop}}}[\frac{1}{1 - p_{\text{drop}}} d \odot h]_i \\
&= \frac{1}{1 - p_{\text{drop}}} \mathbb{E}_{p_{\text{drop}}}[d \odot h]_i \\
&= \frac{1}{1 - p_{\text{drop}}} h_i \mathbb{E}_{p_{\text{drop}}}[d_i] \\
&= \frac{1}{1 - p_{\text{drop}}} h_i (1 - p_{\text{drop}}) \\
&= h_i
\end{aligned}$$

ii

when training dropout is used to prevent overfitting, help all the neurons learning usfull info

at evaluation we want the model's full capacity and a deterministic output, so we use all units with no random dropping; otherwise the same input could give different predictions.

the γ scaling already matches the expected activation, so at test time we just use the full h directly.

(2)

a

stack	buffer	new dependency	transition
[ROOT]	[I, parsed, this, sentence, correctly]		Initial Configu
[ROOT, I]	[parsed, this, sentence, correctly]		SHIFT
[ROOT, I, parsed]	[this, sentence, correctly]		SHIFT
[ROOT, parsed]	[this, sentence, correctly]	parsed→I	LEFT-ARC
[ROOT, parsed, this]	[sentence, correctly]		SHIFT
[ROOT, parsed, this, sentence]	[correctly]		SHIFT
[ROOT, parsed, sentence]	[correctly]	sentence→this	LEFT-ARC
[ROOT, parsed]	[correctly]	parsed→sentence	RIGHT-ARC
[ROOT, parsed, correctly]	[ ]		SHIFT
[ROOT, parsed]	[ ]	parsed→correctly	RIGHT-ARC
[ROOT]	[ ]	ROOT→parsed	RIGHT-ARC

b

$2n$ each of the n words is shifted from buffer to stack once, so n shifts. And each of the n words is removed by an arc action, so n times of arc. in total $2n$

e

Best dev UAS: 88.75; Test UAS: 88.97

f

i
Error type: Verb Phrase Attachment Error
Incorrect dependency: wedding → fearing
Correct dependency: heading → fearing

ii
Error type: Coordination Attachment Error
Incorrect dependency: makes → rescue
Correct dependency: rush → rescue
iii
Error type: Prepositional Phrase Attachment Error
Incorrect dependency: named → Midland
Correct dependency: Joe → Midland

iv
Error type: Modifier Attachment Error
Incorrect dependency: elements $\rightarrow$ most
Correct dependency: crucial $\rightarrow$ most